

Measuring Multilingual Representation Quality

What a controlled study showed, and the metric that came out of it

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The question Aim 1 asks

- How strongly does a model's internal representation encode *which language* a sentence is in, rather than *what it means*?
- Is the mapping between one language's space and another's just linear, or more complex?
- Can we build an intrinsic measure, computed from representations alone, that predicts how a model actually performs?

The third question is the hard one. An intrinsic measure is only worth having if it is right when it matters.

The measure we started with: LFS

A model is given the *same* 300 sentences in 41 languages. Each sentence becomes a vector. Two things drive where those vectors sit:

- **Meaning:** which of the 300 sentences it is
- **Language:** which of the 41 languages it is written in

A variance decomposition separates the two, giving

$$\text{LFS} = \frac{\sigma_{\text{language}}^2}{\sigma_{\text{language}}^2 + \sigma_{\text{meaning}}^2}$$

- Low LFS means the space is organised by meaning, and language identity has been largely factored out. That was taken to be good.

You cannot validate a metric on pretrained models alone

No pretrained model comes with a label saying how much meaning structure it has. So there is nothing to check the metric against. Two ways to create the

missing ground truth:

- **Train models where we control what happened.** 25 variants of Qwen3-0.6B, three seeds each, 75 checkpoints, under different alignment objectives across languages.
- **Distort representations by a known amount.** Take real embeddings, shrink meaning structure by exactly m , or rotate by exactly θ . Now the correct answer follows in closed form.

Every prediction was written down and frozen before the run it referred to.

The data

What the metric reads

NTREX-128: WMT newstest2019, English source professionally translated into 128 languages.

- 1,997 sentences, 123 documents
- 300 used, 41 languages
- Same sentence index means the same meaning in every language
- Splits purged at document level, since neighbouring sentences share context

What it is checked against

Belebele: reading comprehension, 300 questions per language, zero shot.

- All 75 checkpoints sat this exam

Content transfer: does context in another language reduce the model's uncertainty when generating. **Controls:** training data volume per language, language family, script, tokenizer fertility.

What the trained variants did

One variant, trained on the **word level alignment objective named in the proposal**, lost about **95 percent** of the structure encoding meaning at the measurement layer.

And on the metric built to detect exactly that:

LFS	best score in the study
MEXA	third of 25

The measure did not merely miss the damage. It rewarded it.

Why a share cannot see this

$$\text{LFS} = \frac{\sigma_{\text{language}}^2}{\sigma_{\text{language}}^2 + \sigma_{\text{meaning}}^2} \implies \frac{c\sigma_{\text{language}}^2}{c\sigma_{\text{language}}^2 + c\sigma_{\text{meaning}}^2} = \text{unchanged}$$

Shrink both parts together and the share stays where it was.

- Confirmed on controlled distortions: the share moves **less than a millionth** while meaning structure falls **99 percent**
- Inherited by *any* scale invariant ratio, so no re-estimation repairs it
- The concept variance reading, by contrast, returns the injected magnitude exactly: $(1 - m)^2$ to within 0.01

This possibility was written into the July preregistration, with the detector that catches it, before any of these models were trained.

The result nobody predicted

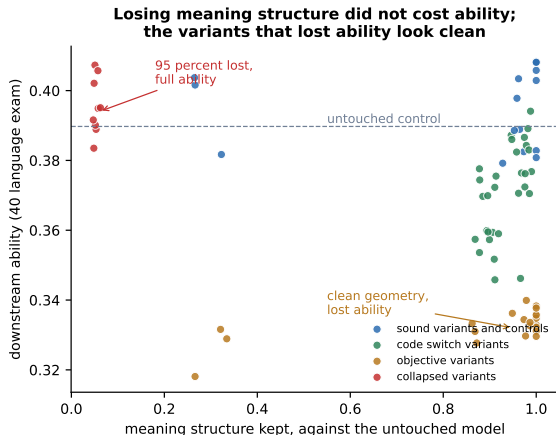
The damaged variant then sat a 40 language reading exam.

damaged variant	0.395
untouched control	0.390

It lost nearly all of its meaning structure and paid nothing for it, in every seed.

The variants that *did* lose ability, by roughly six points, are the ones whose geometry looks clean.

Representational damage and capability damage are separable. That was not the working assumption.



So what should a metric report?

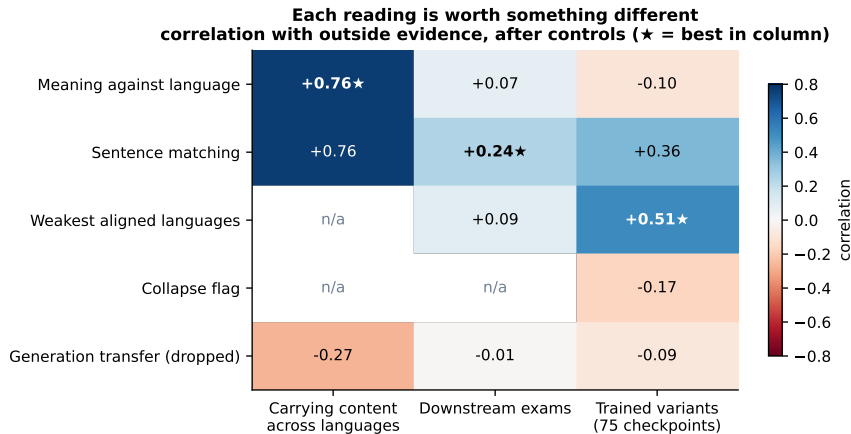
If one number can hide a 95 percent loss, and if geometric health does not imply capability, then no single score is safe.

RMFS reports four readings per language, side by side.

- **Meaning against language** — LFS's original decomposition, kept intact
- **Sentence matching** — can the model find a sentence's translation among distractors
- **Weakest aligned languages** — behaviour in the poorest aligned tail, where training damage appears first
- **Collapse flag** — meaning structure against a reference checkpoint, for training settings

They are kept apart because each one predicts something the others miss. The next slide is that claim, measured.

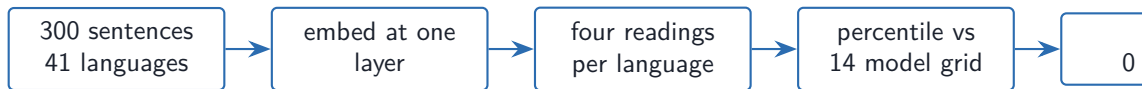
Each reading is worth something different



Correlations after removing training data volume, language family, script and tokenizer effects.

No reading wins twice — which is the argument against averaging them into one score.

How a score is computed



- About **four GPU minutes** per model
- Each reading is converted to its percentile against a frozen reference grid of 14 models, then averaged
- A score of **74** means better than 74 percent of the reference models
- Reported per language; the model score is the median across languages

Percentiles rather than raw values, because the four readings live on different scales and averaging raw scales is what hides damage.

The scoreboard

Qwen3-8B	90.5	Falcon3-7B	50.0
Qwen3-4B	85.7	OLMo-2-7B	50.0
Qwen3-1.7B	73.8	EuroLLM-1.7B	45.2
Salamandra-7B	69.0	bloom-1b7	42.9
bloom-7b1	54.8	SmolLM2-1.7B	35.7
Qwen3-0.6B	54.8	Mistral-7B	33.3
		OLMo-2-1B	28.6

- Scale ordering within a family falls out with no tuning
- Multilingual focused models sit above English centric ones of the same size

Is the mapping between languages linear?

Across **1,792** model and language pairs, fitting a ladder of maps from simplest to most complex and selecting the simplest that is not beaten:

linear or simpler	74 percent
linear exactly	60 percent
needs a nonlinear map	26 percent

- The nonlinear cases are **concentrated, not spread**: three models (Falcon3-7B, OLMo-2-1B, OLMo-2-7B) carry 80 percent of them
- Across the other eleven models, only **6.5 percent** of languages need a nonlinear map

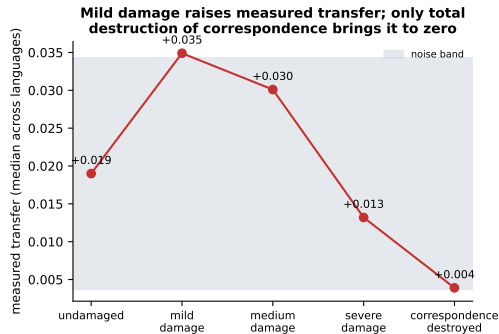
An answer to 3.1.1 as a distribution rather than a single case.

A measure built on generation can run backwards

A transfer measure taken from the generation harness *rises* under mild damage, and falls only when the correspondence between languages is destroyed outright.

- Seen in trained variants
- Reproduced in dose controlled distortions
- Confirmed across models

Impoverished representations lean harder on whatever context they are given, so a measure that injects context rewards the damage it should detect. This reading was dropped from the score.



What the numbers rest on

- **Downstream prediction sits near 0.3 after controls** — for every measure in this area, ours included. Raw correlations are much higher, and are mostly training data volume.
- **The 0.76 holds under language family resampling** (0.60 to 0.89) but rests on four models in that comparison: treating models as the unit gives 0.12 to 0.84.
- **A panel of 69 languages carries the independent evidence of roughly two.** Languages are so mutually correlated that nominal sample size overstates the evidence by an order of magnitude.

The third point bounds what any measure in this area can claim from per language correlations, including this one.

How the work was checked

Every prediction recorded before its data, and how it scored

Collapsed variant ranks last on the single score	FAIL
Detect collapse across the whole signature table	FAIL
Collapse detector exact at every level tested	PASS
Unchanged under a shared rotation	PASS
Respond to offsets, rotations and warps	FAIL
Predicts how content carries across languages	PASS
Same, excluding the generation measure	PASS
Downstream signal on the frozen model set	PASS
Same signal across the full grid	FAIL
Adds to the strongest published measure	BRANCH
Collapse costs at least 5 points of ability	FAIL
Published measure misranks this panel	VOID
Panel validity given the above	VOID
Alignment training adds no downstream gain	PASS

- Every prediction was recorded and frozen **before** the data it referred to existed
- Distortions with closed form answers set the tolerances
- Correlations are confound controlled, with intervals from resampling whole languages
- Validity was also measured on a panel that **contains** the damaged models

The failures are on the slide with the passes.

Against Aim 1's questions

3.1.1 strength of the language component; linear or more complex	Quantified across 14 models. Linear or simpler in 74 percent of model and language pairs, with nonlinear cases concentrated in three models.
3.1.2 logit lens	Covered in the earlier phase; carried as context.
3.1.3 transfer in generation	Harness built and validated; control patches reproduce the reference exactly. Supplies one of the three kinds of outside evidence, and produced the inversion result.
3.1.4 intrinsic measures that correlate with extrinsic ones	Measured against three kinds of outside evidence with confound controls, and the same analysis bounds how much independent evidence a language panel carries.

